GLOBAL URBAN HEAT

Persistent-city accumulation reflects turnover and duration-dependent exit

23 annual transitions / 6,081 high-state entries / 4,726 exits

A Cities move between three states

Exits below threshold

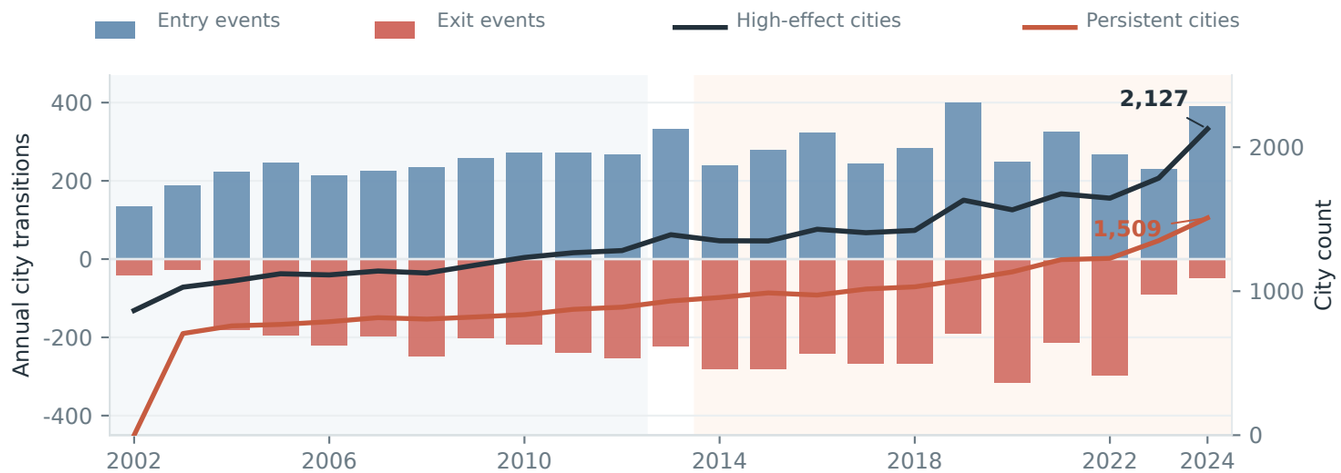
After 1–2 years 3,679

After ≥3 years 1,047

2024 city counts · High-state flows: 2001–2024

Persistent flows: 2003–2024 · Initial set: 706 · Net gain: +803

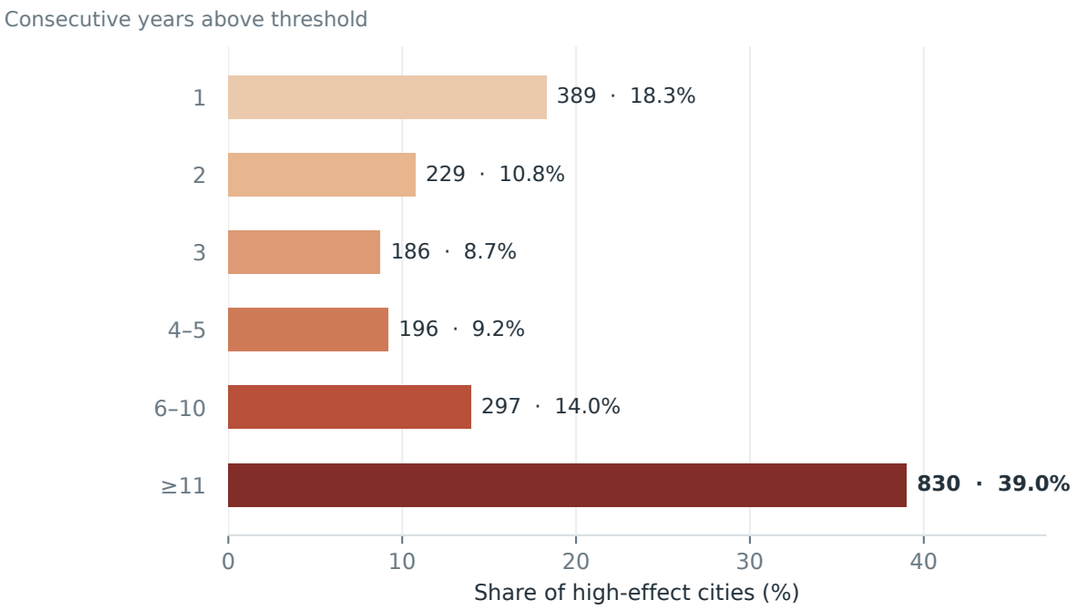
B Turnover persisted as the city pool expanded



Entry 2.66% · Exit 16.92%

Entry 3.57% · Exit 14.94%

C The duration structure in 2024



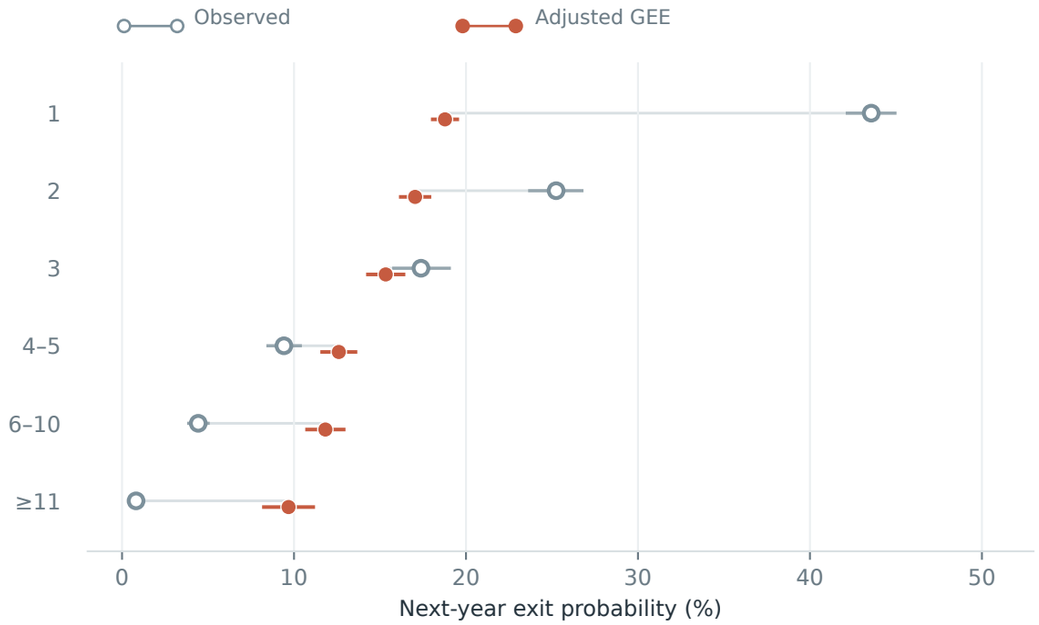
2,127

high-effect cities

39.0%

high for ≥11 years

D Longer duration is associated with lower exit



Adjusted OR per doubling 0.735

95% CI 0.702–0.770 · 2,721 cities · 29,853 at-risk city-years

A: 2024 stocks, historical flows. B: pooled eligible risk records. C: 2024 snapshot. D: retrospective GEE; 95% city-cluster bootstrap (observed) and delta-method (adjusted) intervals.